

Evidence-guided conservative graph calibration for direct semi-supervised clustering

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Abstract

Graph-based semi-supervised clustering usually relies on a prescribed neighborhood graph, although limited labeled data can also provide task-specific evidence about whether that graph should be modified. We formulate graph modification as an evidence-guided Conservative Graph Calibration (CGC) problem for Laplacian-regularized semi-supervised representation models. A binary baseline graph and a same-support locally weighted candidate are evaluated by out-of-fold harmonic label propagation. Their foldwise predictive-loss difference estimates the advantage of local weighting, while its foldwise variability discounts unstable evidence. The resulting retained-advantage coefficient determines how far the final graph moves toward the candidate and yields exact baseline recovery when the evidence for modification is insufficient. CGC is completed before downstream optimization and introduces neither a graph-learning variable nor an additional objective coefficient. We first instantiate CGC in one-hot-constrained NMF. Across six benchmarks, CGC-GOCNMF improves its parent on COIL20, COIL100, Optdigit-s, and MNIST while retaining essentially baseline behavior on PIE and YaleB. We then transfer the same calibration rule, without retuning it, to GNMFLD, whose supervision uses a label-discrimination penalty rather than fixed one-hot rows. CGC-GNMFLD improves ACC and NMI in all 20 paired trials on each benchmark, with mean ACC gains of 0.24-4.57 percentage points and NMI gains of 0.57-3.06 points. Industrial-image, representation-robustness, ablation, sensitivity, and controlled-corruption experiments further characterize when candidate weighting is retained, attenuated, or rejected. These results support CGC as a reusable pre-optimization mechanism for regulating structural graph modification from limited label evidence.

Keywords: direct semi-supervised clustering; nonnegative matrix factorization; conservative graph calibration; label evidence; graph regularization; variability-adjusted shrinkage; steel surface defect clustering

1 Introduction

Nonnegative matrix factorization (NMF) decomposes a non-negative data matrix into two low-rank nonnegative factors and provides an interpretable representation of high-dimensional data [1, 2]. Owing to its parts-based representation and simple optimization structure, NMF has been widely used for dimensionality reduction, image representation, and clustering. Standard NMF, however, is unsupervised and does not directly exploit either the local geometry of the data or the labels available for a small subset of samples.

Graph-regularized NMF (GNMF) incorporates manifold information by encouraging neighboring samples to have similar low-dimensional representations [3]. Semi-supervised NMF further uses partial labels through hard constraints, label-discrimination losses, or propagated supervisory information [4, 5, 6]. More elaborate approaches learn multiple graph weights, update neighborhood structures jointly with the factors, or combine NMF with label propagation and hypergraph learning [7, 8, 9, 10, 11, 12]. These methods can improve clustering, but they often introduce additional variables, regularization coefficients, or iterative subproblems. Many NMF clustering methods also require K -means or another post-processing procedure, so the final assignments are not obtained directly from the factorization.

Graph regularized one-hot-constrained NMF (GOCNMF) provides a concise alternative for direct semi-supervised clustering [13]. It fixes the representations of labeled sam-

ples as one-hot vectors and uses graph regularization to guide the unlabeled representations toward class-indicative coordinates. The cluster assignment of each sample is therefore obtained from the largest coordinate of its representation, without an external classifier or K -means. This architecture preserves the simplicity of NMF and avoids an additional label-fitting coefficient.

The reliability of graph regularization nevertheless depends on the prescribed neighborhood structure. A binary p -nearest-neighbor graph is simple and stable, but it ignores variations in local similarity strength. A locally scaled graph can encode such variations more faithfully [14], yet greater expressiveness does not necessarily imply greater task relevance. In a semi-supervised setting, the available labels provide direct evidence about whether changing the graph improves class-consistent propagation. This leads to a distinct methodological problem: graph modification should itself be treated as a decision under limited supervisory evidence rather than as an unconditional preprocessing choice.

This perspective assigns the labeled samples two complementary roles. First, they supervise the downstream representation according to the parent model, for example through fixed one-hot rows or a label-discrimination loss. Second, they provide task-specific evidence for deciding how far the graph should depart from a trusted baseline. The second role is usually implicit in fixed-graph methods or absorbed into a larger joint graph-learning model. Here it is treated explicitly through a pre-factorization calibration mechanism.

m. We therefore study whether limited labeled evidence can regulate structural graph modification before representation learning, and whether instability in the observed graph advantage can be used to control the amount of deviation from the baseline.

To address this problem, we develop an evidence-guided Conservative Graph Calibration (CGC) mechanism. CGC compares a prescribed baseline graph $W^{(0)}$ with a same-support candidate graph $W^{(1)}$ using out-of-fold harmonic label propagation [15] and multiclass squared prediction loss [16]. If Δ denotes the mean foldwise advantage of the candidate graph and s its foldwise variability, the retained-advantage coefficient is

$$\theta = \begin{cases} \frac{(\Delta - s)_+}{\Delta}, & \Delta > 0, \\ 0, & \Delta \leq 0, \end{cases}$$

with $W_\theta = (1 - \theta)W^{(0)} + \theta W^{(1)}$. Thus, only the fraction of the observed advantage that survives the variability discount is allowed to modify the graph. When $\Delta \leq s$, the coefficient is zero and the baseline is recovered exactly.

We first instantiate CGC in GOCNMF, yielding CGC-GOCNMF. Calibration is completed before factorization and is not optimized jointly with the factor matrices, so the fixed one-hot constraints, multiplicative solver, and direct row-wise arg max assignment rule of the parent are retained. To test whether the same calibration mechanism depends on this hard one-hot supervision, we further transfer CGC to GNMFLD [6], which instead uses a label-discrimination penalty. In both instantiations, CGC changes only the graph supplied to the parent optimizer.

The main contributions are summarized as follows.

1. **Evidence-guided conservative graph calibration.** We formulate graph modification as a pre-optimization decision for Laplacian-regularized semi-supervised representation models. A baseline graph and a same-support candidate are evaluated from out-of-fold labeled predictions rather than unconditionally replacing or jointly learning the graph.
2. **Variability-adjusted retained advantage.** CGC discounts a positive mean graph advantage by its fold-to-fold variability. The resulting coefficient has an explicit retained-advantage interpretation, is monotone in the evidence and variability terms, and recovers the baseline exactly when the evidence for graph modification is insufficient.
3. **Controlled structural perturbation and parent-model preservation.** We show that the calibration coefficient scales the graph and Laplacian deviations exactly and bounds the induced change in any downstream objective of the stated Laplacian-regularized form. For the primary GOCNMF instantiation, we additionally establish graph validity, monotonic objective values under the exact multiplicative updates, and stationarity of strictly positive fixed points.
4. **Two-parent empirical validation.** The primary CGC-GOCNMF study combines six benchmarks, an indus-

trial image dataset, representation robustness, ablation, sensitivity analysis, controlled graph corruption, and timing. A second controlled transfer to GNMFLD uses the same calibration rule with no retuning and tests whether CGC remains effective under a distinct label-discrimination supervision mechanism.

The remainder of this paper is organized as follows. Section 2 reviews related semi-supervised and graph-learning NMF methods. Section 3 introduces the candidate graphs, out-of-fold assessment, and conservative calibration rule. Section 4 presents the structural, convergence, and complexity analyses. Section 5 reports the experimental results. Section 6 concludes the paper.

2 Related work

2.1 Semi-supervised NMF

Semi-supervised NMF uses limited labels through hard constraints, discriminative losses, propagated supervision, or self-generated supervisory structure. Representative examples include CNMF [4], GNMFLD [6], robust and dual-supervision variants [17, 18], self-supervised semi-supervised NMF [19], EDNMF [20], and methods based on label or pairwise-constraint propagation [5, 7, 10]. Deep semi-supervised graph clustering further combines nonlinear representation learning with graph-based supervision [21].

These methods primarily change how supervision enters representation learning. CGC addresses the complementary structural question of whether a prescribed graph should be modified before the downstream model is optimized. Harmonic propagation is used only to compare candidate graphs from held-out labeled scores; no pseudo-label matrix is inserted into the downstream objective.

2.2 Multiple and adaptive graph learning

GNMF introduced Laplacian regularization to preserve local manifold geometry [3]. Later work learned multiple-graph weights, adaptive neighbors, or higher-order structures jointly with the representation, including MSNMF [12], AGSS-NMF [11], ABNMTF [8], and hypergraph variants [9, 22]. These approaches optimize graph-related quantities inside the learning model.

CGC instead compares two fixed same-support graphs before downstream optimization. Held-out label-prediction quality determines the direction of graph modification, while foldwise variability contracts unstable evidence. This differs from hard validation-based graph selection, grid search over mixture weights, and joint adaptive-graph learning. It is also distinct from safety-aware graph-based semi-supervised learning, where graph selection is coupled with a safety-oriented learner [23]: CGC regulates how far a prescribed graph is allowed to move before the parent representation model is optimized. The variability term is therefore used as a deterministic stability discount rather than as a statistical confidence bound.

Table 1: Methodological positioning of representative semi-supervised NMF methods.

Method	Label use	Graph handling	Decision stage	Direct	Recovery
GNMFLD	Discrimination loss	Fixed graph	Pre-factorization	Yes	–
CGC-GNMFLD	Same + OOF evidence	Calibrated fixed pair	Pre-factorization	Yes	Exact
EDNMF	Element constraints	No graph	–	Yes	–
MSNMF	Pairwise constraints	Adaptive multi-graph	Joint	No	No
ABNMTF	Label block structure	Adaptive neighbors	Joint	No	No
GOCNMF	Fixed one-hot rows	Binary p -NN	Pre-factorization	Yes	–
CGC-GOCNMF	One-hot + OOF evidence	Calibrated fixed pair	Pre-factorization	Yes	Exact

GNMFLD and GOCNMF use fixed graphs, whereas MSNMF and ABNMTF learn graph-related quantities during optimization. The two CGC instantiations retain their respective parent supervision mechanisms and replace only the prescribed graph by an evidence-calibrated convex combination of two fixed same-support candidates; when retained evidence vanishes, the parent graph is recovered exactly.

2.3 Direct clustering with one-hot constraints

Many NMF-based clustering methods require K -means or another post-processing procedure to obtain the final assignments. GOCNMF instead fixes the labeled representations as one-hot vectors and directly assigns each unlabeled sample according to its largest representation coordinate [13]. This avoids an external classifier and an additional label-fitting coefficient.

CGC-GOCNMF preserves the factorization structure, one-hot constraints, and direct assignment rule of GOCNMF while introducing a separate evidence-guided decision for graph modification before factorization. The transfer instantiation CGC-GNMFLD preserves GNMFLD’s label-discrimination term and direct class-coordinate assignment instead. In both cases, the calibration coefficient is determined from out-of-fold labeled predictions before factorization; when $\Delta \leq s$, it becomes zero and the original parent graph is recovered exactly.

3 Evidence-guided conservative graph calibration

3.1 General calibration setting

Let

$$X = [x_1, \dots, x_n] \in \mathbb{R}_+^{m \times n}$$

be a nonnegative data matrix whose columns are normalized to unit Euclidean norm. The samples belong to c classes. Let $\mathcal{L}$ and $\mathcal{U}$ denote the labeled and unlabeled index sets, respectively, where $\mathcal{L} \cap \mathcal{U} = \emptyset$ and $\mathcal{L} \cup \mathcal{U} = \{1, \dots, n\}$. For $i \in \mathcal{L}$, let $y_i \in \{1, \dots, c\}$ be the class label and $e_{y_i} \in \{0, 1\}^c$ the corresponding one-hot vector. Denote the matrix of labeled one-hot vectors by

$$C_{\mathcal{L}} = [e_{y_i}^T]_{i \in \mathcal{L}}.$$

CGC is defined before the downstream representation model is optimized. It takes a baseline graph $W^{(0)}$, a candidate graph $W^{(1)}$, and the labeled subset as input; it then uses held-out label evidence to construct a calibrated graph W_{θ} . To make the scope of this mechanism explicit, consider

a generic graph-regularized objective

$$\mathcal{J}_L(Z) = \Phi(Z) + \gamma \text{Tr}(H(Z)^T L H(Z)), \quad \gamma > 0, \quad (1)$$

where Z denotes the model variables, Φ contains the graph-independent part of the objective, and $H(Z) \in \mathbb{R}^{n \times r}$ is the representation on which graph regularization acts. CGC modifies only the Laplacian supplied to (1); it does not alter Φ , introduce a graph variable into Z , or prescribe a particular downstream solver. Graph-regularized NMF is a direct instance with $H(Z) = V$. Sections 3.5 and 3.6 give two downstream instantiations with different supervision mechanisms.

3.2 Binary and locally scaled graphs

For the normalized samples, define the cosine distance

$$d_{ij} = \max\{0, 1 - x_i^T x_j\}. \quad (2)$$

Let $\mathcal{N}_p(i) \subseteq \{1, \dots, n\} \setminus \{i\}$ contain the p nearest neighbors of sample i , and define the undirected support

$$\mathcal{E}_p = \{\{i, j\} : 1 \leq i < j \leq n, j \in \mathcal{N}_p(i) \text{ or } i \in \mathcal{N}_p(j)\}.$$

The binary graph is

$$w_{ij}^{(0)} = \begin{cases} 1, & \{i, j\} \in \mathcal{E}_p, \\ 0, & \text{otherwise,} \end{cases} \quad w_{ii}^{(0)} = 0. \quad (3)$$

Let i_p be the p -th nearest neighbor of i , and set

$$\sigma_i = \max\{d_{i, i_p}, \varepsilon\}. \quad (4)$$

Following the local-scaling principle [14], define

$$a_{ij} = \begin{cases} \exp\left[-\left(\frac{d_{ij}}{\sqrt{\sigma_i \sigma_j} + \varepsilon}\right)^2\right], & j \in \mathcal{N}_p(i), \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Set

$$\hat{w}_{ij}^{(1)} = \max\{a_{ij}, a_{ji}\}, \quad \bar{w} = \frac{1}{|\mathcal{E}_p|} \sum_{\{r, s\} \in \mathcal{E}_p} \hat{w}_{rs}^{(1)},$$

and define

$$w_{ij}^{(1)} = \begin{cases} \hat{w}_{ij}^{(1)} / \bar{w}, & \{i, j\} \in \mathcal{E}_p, \\ 0, & \text{otherwise,} \end{cases} \quad w_{ii}^{(1)} = 0. \quad (6)$$

Therefore, $W^{(0)}$ and $W^{(1)}$ are symmetric, have identical off-diagonal support, and differ only in their edge weights.

3.3 Out-of-fold graph assessment

Partition the labeled set into F nonempty approximately stratified folds,

$$\mathcal{L} = \mathcal{L}_1 \dot{\cup} \dots \dot{\cup} \mathcal{L}_F.$$

Here, approximate stratification balances the available labeled samples from each class across the folds as evenly as their integer counts permit. It does not require every class to occur in every held-out fold. This distinction is necessary when a class contains fewer than F labeled samples. Each labeled sample is held out exactly once, and each training set retains the remaining labeled samples from that class whenever the class contains at least two labeled observations. For fold f , define

$$\mathcal{T}_f = \mathcal{L} \setminus \mathcal{L}_f, \quad \mathcal{Q}_f = \{1, \dots, n\} \setminus \mathcal{T}_f.$$

For $q \in \{0, 1\}$, let

$$L^{(q)} = D^{(q)} - W^{(q)}, \quad D^{(q)} = \text{diag}(W^{(q)} \mathbf{1}).$$

The harmonic label scores $H_f^{(q)} \in \mathbb{R}^{|\mathcal{Q}_f| \times c}$ satisfy

$$L_{\mathcal{Q}_f \mathcal{Q}_f}^{(q)} H_f^{(q)} = W_{\mathcal{Q}_f \mathcal{T}_f}^{(q)} C_{\mathcal{T}_f}, \quad (7)$$

where $C_{\mathcal{T}_f}$ contains the one-hot labels in $\mathcal{T}_f$. If MATLAB reports a singular system or the solution contains nonfinite entries, the stabilized system with $10^{-10}I$ is solved. Negative roundoff values are truncated to zero, and each row with positive sum is normalized. A row with zero sum is retained as the zero score vector. Such a vector is not interpreted as a probability distribution; in (8) it receives unit squared loss against any one-hot target and therefore penalizes a candidate graph that cannot propagate labeled information to that observation. Denote the resulting score vector for $i \in \mathcal{L}_f$ by $p_{f,i}^{(q)}$. The foldwise squared prediction loss is

$$R_f^{(q)} = \frac{1}{|\mathcal{L}_f|} \sum_{i \in \mathcal{L}_f} \|p_{f,i}^{(q)} - e_{y_i}\|_2^2. \quad (8)$$

Define

$$\delta_f = R_f^{(0)} - R_f^{(1)}, \quad \Delta = \frac{1}{F} \sum_{f=1}^F \delta_f, \quad (9)$$

and

$$s = \left[\frac{1}{F(F-1)} \sum_{f=1}^F (\delta_f - \Delta)^2 \right]^{1/2}. \quad (10)$$

Thus, $\Delta > 0$ favors the locally weighted graph, whereas s summarizes the fold-to-fold variability of this advantage. It is used as a variability measure rather than a confidence bound. All held-out predictions use labels from $\mathcal{T}_f$ only; no ground-truth label from $\mathcal{U}$ is used.

3.4 Variability-adjusted conservative calibration

To make the role of foldwise variability explicit, consider the shrinkage family:

$$\theta_\kappa = \begin{cases} \frac{(\Delta - \kappa s)_+}{\Delta}, & \Delta > 0, \\ 0, & \Delta \leq 0, \end{cases} \quad \kappa \geq 0, \quad (11)$$

where $(a)_+ := \max\{a, 0\}$. Here, κ controls how strongly the foldwise variability is discounted. When $\kappa = 0$, $\theta_\kappa = 1$ for every $\Delta > 0$, which is the hard graph-selection rule. For $\kappa > 0$, a positive advantage is retained only to the extent that it exceeds κs , and exact fallback occurs whenever $\Delta \leq \kappa s$. Both formal CGC instantiations use $F = 5$ and $\kappa = 1$, so

$$\theta := \theta_1 = \begin{cases} \frac{(\Delta - s)_+}{\Delta}, & \Delta > 0, \\ 0, & \Delta \leq 0. \end{cases} \quad (12)$$

Thus, $(\Delta - \kappa s)_+$ is the variability-adjusted retained advantage, and θ_κ is the fraction of the observed advantage that remains after the stability discount. Section 5 examines $\kappa \in \{0, 0.5, 1, 1.5\}$ and $F \in \{3, 5, 10\}$.

Proposition 1 (Basic properties of the calibration coefficient). *Let $s \geq 0$, $\kappa \geq 0$, and let θ_κ be defined by (11). Then $0 \leq \theta_\kappa \leq 1$. For fixed $\Delta > 0$, θ_κ is nonincreasing in s and κ ; for fixed s and κ , it is nondecreasing in $\Delta > 0$. Moreover, for fixed $\Delta > 0$,*

$$|\theta_\kappa(s_1) - \theta_\kappa(s_2)| \leq \frac{\kappa}{\Delta} |s_1 - s_2|.$$

In particular, $\theta_\kappa = 0$ whenever $\Delta \leq \kappa s$.

Proof. For $\Delta > 0$,

$$\theta_\kappa = \max\left\{1 - \frac{\kappa s}{\Delta}, 0\right\}.$$

The range and monotonicity statements follow directly. Since the positive-part map is 1-Lipschitz, the stated bound follows immediately. For $\Delta \leq 0$, $\theta_\kappa = 0$ by definition. $\square$

The calibrated graph is

$$W_\theta = (1 - \theta)W^{(0)} + \theta W^{(1)}. \quad (13)$$

Because $0 \leq \theta \leq 1$, this is a convex combination of two symmetric nonnegative graphs with identical support. Let

$$D_\theta = \text{diag}(W_\theta \mathbf{1}), \quad L_\theta = D_\theta - W_\theta.$$

Proposition 2 (Controlled structural deviation). *For every $\theta \in [0, 1]$,*

$$\begin{aligned} W_\theta - W^{(0)} &= \theta(W^{(1)} - W^{(0)}), \\ L_\theta - L^{(0)} &= \theta(L^{(1)} - L^{(0)}). \end{aligned}$$

Consequently, for any absolutely homogeneous matrix norm,

$$\begin{aligned} \|W_\theta - W^{(0)}\| &= \theta \|W^{(1)} - W^{(0)}\|, \\ \|L_\theta - L^{(0)}\| &= \theta \|L^{(1)} - L^{(0)}\|. \end{aligned}$$

Proof. The first identity follows from (13). Because the degree operator is linear,

$$D_\theta = (1 - \theta)D^{(0)} + \theta D^{(1)},$$

and therefore $L_\theta = (1 - \theta)L^{(0)} + \theta L^{(1)}$. The norm equalities follow from $\theta \geq 0$ and absolute homogeneity. $\square$

3.5 Primary instantiation in one-hot-constrained NMF

We now instantiate CGC in GOCNMF. The parent model solves

$$\min_{U, V_{\mathcal{U}:} \geq 0} \|X - UV^T\|_F^2 + \alpha \text{Tr}(V^T L^{(0)} V),$$

$$V_{\mathcal{L}:} = C_{\mathcal{L}}. \quad (14)$$

where $U \in \mathbb{R}_+^{m \times c}$, $V \in \mathbb{R}_+^{n \times c}$, and $\alpha > 0$. The unlabeled samples are assigned directly by

$$\hat{y}_i = \arg \max_{1 \leq k \leq c} v_{ik}, \quad i \in \mathcal{U}. \quad (15)$$

For this model, (1) has $Z = (U, V)$, $H(Z) = V$, $\gamma = \alpha$, and $\Phi(U, V) = \|X - UV^T\|_F^2$ subject to the one-hot labeled rows. Replacing $L^{(0)}$ by L_θ gives CGC-GOCNMF:

$$\min_{U, V_{\mathcal{U}:} \geq 0} J_\theta(U, V) = \|X - UV^T\|_F^2 + \alpha \text{Tr}(V^T L_\theta V),$$

$$V_{i:} = e_{y_i}^T, \quad i \in \mathcal{L}. \quad (16)$$

Proposition 3 (Controlled downstream objective perturbation). *Let $\mathcal{J}_L$ be any objective of the form (1). For every feasible Z ,*

$$\mathcal{J}_{L_\theta}(Z) - \mathcal{J}_{L^{(0)}}(Z) = \gamma \theta \text{Tr}[H(Z)^T (L^{(1)} - L^{(0)}) H(Z)],$$

and therefore

$$|\mathcal{J}_{L_\theta}(Z) - \mathcal{J}_{L^{(0)}}(Z)| \leq \gamma \theta \|L^{(1)} - L^{(0)}\|_2 \|H(Z)\|_F^2.$$

Proof. The graph-independent term $\Phi(Z)$ cancels. Proposition 2 gives $L_\theta - L^{(0)} = \theta(L^{(1)} - L^{(0)})$, which yields the identity. Since $L^{(1)} - L^{(0)}$ is symmetric,

$$\left| \text{Tr}[H^T (L^{(1)} - L^{(0)}) H] \right| \leq \|L^{(1)} - L^{(0)}\|_2 \|H\|_F^2.$$

This proof is complete. $\square$

The calibration coefficient is fixed before this optimization and is not updated with U or V . Thus, CGC changes only the Laplacian supplied to the parent objective: the one-hot constraints, nonnegativity conditions, factorization variables, and direct assignment rule remain unchanged. The choice $\kappa = 1$ belongs to the calibration stage rather than to the NMF regularization parameters.

For fixed W_θ , the multiplicative updates are

$$U \leftarrow U \odot \frac{XV}{U(V^T V)} \quad (17)$$

and

$$V_{\mathcal{U}:} \leftarrow V_{\mathcal{U}:} \odot \frac{(X^T U + \alpha W_\theta V)_{\mathcal{U}:}}{(V(U^T U) + \alpha D_\theta V)_{\mathcal{U}:}}, \quad V_{\mathcal{L}:} = C_{\mathcal{L}}. \quad (18)$$

Fractions and products are entrywise where appropriate. The analysis in Section 4 concerns the exact multiplicative updates under the strict-positivity condition stated in Theorem 3. In the implementation, denominators are lower bounded by ε for numerical protection.

Algorithm 1. CGC instantiated in GOCNMF (CGC-GOCNMF)

Input: X , labeled set $\mathcal{L}$, labels $\{y_i : i \in \mathcal{L}\}$, number of classes c , neighbors p , graph coefficient α , folds $F = 5$, maximum iterations T .

Output: U, V , calibration coefficient θ , and assignments $\hat{y}$.

1. Normalize each column of X to unit Euclidean norm.
2. Construct $W^{(0)}$ from (3) and $W^{(1)}$ from (4)-(6).
3. Divide $\mathcal{L}$ into $F = 5$ nonempty approximately stratified folds, balancing each class as evenly as its labeled count permits.
4. For $q = 0, 1$ and $f = 1, \dots, F$, hide $\mathcal{L}_f$, solve the harmonic propagation problem, and compute $R_f^{(q)}$.
5. Compute δ_f , Δ , and s , set $\kappa = 1$, and obtain θ and W_θ from (12)-(13).
6. Initialize positive U and $V_{\mathcal{U}:}$, and fix $V_{\mathcal{L}:} = C_{\mathcal{L}}$.
7. Repeat (17) and (18) for T iterations.
8. Return $\hat{y}_i = \arg \max_{1 \leq k \leq c} v_{ik}$ for $i \in \mathcal{U}$.

3.6 Second downstream instantiation in GNMFLD

To test whether CGC depends on the hard one-hot rows of GOCNMF, we also instantiate it in GNMFLD [6]. Let $Y \in \mathbb{R}_+^{n \times c}$ contain the available labels, with $Y_{i:} = e_{y_i}^T$ for $i \in \mathcal{L}$ and $Y_{i:} = 0$ otherwise, and let $\mathcal{P}_{\mathcal{L}}$ retain the rows indexed by $\mathcal{L}$ and set all remaining rows to zero. The original GNMFLD label-discrimination term is $\|\mathcal{P}_{\mathcal{L}}(V) - \mathcal{P}_{\mathcal{L}}(Y)\|_F^2$; here $\mathcal{P}_{\mathcal{L}}(Y) = Y$ by construction. The parent objective is therefore

$$\min_{U, V \geq 0} \|X - UV^T\|_F^2 + \alpha \|\mathcal{P}_{\mathcal{L}}(V) - \mathcal{P}_{\mathcal{L}}(Y)\|_F^2$$

$$+ \beta \text{Tr}(V^T L^{(0)} V). \quad (19)$$

where $\alpha, \beta > 0$. Thus α weights label discrimination and β weights graph regularization. Unlike GOCNMF, the labeled rows of V are not fixed; they are encouraged toward their class indicators by the discrimination term.

CGC-GNMFLD replaces only $L^{(0)}$ by L_θ :

$$\min_{U, V \geq 0} J_\theta^{\text{LD}}(U, V) = \|X - UV^T\|_F^2$$

$$+ \alpha \|\mathcal{P}_{\mathcal{L}}(V) - \mathcal{P}_{\mathcal{L}}(Y)\|_F^2$$

$$+ \beta \text{Tr}(V^T L_\theta V). \quad (20)$$

Thus (1) applies with $H(Z) = V$ and $\gamma = \beta$. Proposition 3 gives

$$|J_\theta^{\text{LD}}(U, V) - J_0^{\text{LD}}(U, V)| \leq \beta \theta \|L^{(1)} - L^{(0)}\|_2 \|V\|_F^2.$$

For a fixed calibrated graph, the parent multiplicative updates are retained:

$$U \leftarrow U \odot \frac{XV}{U(V^T V)},$$

$$V \leftarrow V \odot \frac{X^T U + \alpha Y + \beta W_\theta V}{V(U^T U) + \alpha \mathcal{P}_\mathcal{L}(V) + \beta D_\theta V}.$$

where $\mathcal{P}_\mathcal{L}(Y) = Y$. The unlabeled assignments remain $\hat{y}_i = \arg \max_k v_{ik}$. No additional coefficient is introduced: the transfer uses the same $F = 5$, $\kappa = 1$ calibration rule and changes only the graph supplied to the GNMFLD solver.

4 Structural properties and primary-instantiation convergence

4.1 Validity and support preservation

Proposition 4 (Support preservation). *The two candidate graphs satisfy*

$$w_{ij}^{(0)} > 0 \iff w_{ij}^{(1)} > 0.$$

Consequently, W_θ has the same off-diagonal support for every $\theta \in [0, 1]$.

Proof. Every directed nearest-neighbor edge receives a strictly positive exponential weight in (5). Max-symmetrization therefore gives a positive entry exactly on the union of the directed neighbor relations, which is $\mathcal{E}_p$. Rescaling by the positive mean $\bar{w}$ does not change the support. A convex combination of two positive entries remains positive, while two zero entries remain zero. This completes the proof. $\square$

Theorem 1 (Graph validity). *For every $\theta \in [0, 1]$, W_θ is symmetric and nonnegative, and L_θ is positive semidefinite.*

Proof. Both candidate graphs are symmetric and nonnegative, so their convex combination has the same properties. For any $z \in \mathbb{R}^n$,

$$z^T L_\theta z = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n w_{\theta,ij} (z_i - z_j)^2 \geq 0.$$

Thus $L_\theta \succeq 0$. The proof is complete. $\square$

4.2 Exact fallback

Theorem 2 (Exact baseline recovery). *If $\Delta \leq s$, then $\theta = 0$, $W_\theta = W^{(0)}$, and $L_\theta = L^{(0)}$. Consequently, every downstream model of the form (1) exactly recovers its baseline graph-regularized objective.*

Proof. If $\Delta \leq 0$, the second branch of (12) gives $\theta = 0$. If $0 < \Delta \leq s$, then $1 - s/\Delta \leq 0$, so the first branch also gives $\theta = 0$. Equation (13) yields $W_\theta = W^{(0)}$, and linearity of the degree operator gives $L_\theta = L^{(0)}$. Substitution into (1) proves exact baseline recovery. $\square$

Corollary 1 (Exact recovery of the instantiated parents). *Under the condition of Theorem 2, CGC-GOCNMF and CGC-GNMFLD exactly recover their respective parent objectives. With the same labeled split, initialization, solver, and iteration budget, the corresponding multiplicative updates, learned representations, and direct assignments also coincide.*

Remark 1. Exact recovery is a property of the calibration stage itself and is independent of whether the parent uses hard one-hot rows or a label-discrimination penalty.

4.3 Monotonicity of the primary GOCNMF instantiation

Theorem 3 (Monotonic objective values). *Assume that W_θ is fixed before factorization and that, throughout the iterations, all free entries of U and $V_{\mathcal{U}}$, together with the denominators in (17)-(18), remain strictly positive. With $V_{\mathcal{L}}$ kept fixed, one complete alternating iteration satisfies*

$$J_\theta(U^{(t+1)}, V^{(t+1)}) \leq J_\theta(U^{(t)}, V^{(t)}).$$

Consequently, the sequence of objective values converges.

Proof. We first use the following elementary inequality. If M is symmetric and entrywise nonnegative, $z > 0$, and h is any real vector, then

$$h^T M h \leq \sum_k \frac{(Mz)_k}{z_k} h_k^2. \quad (21)$$

Indeed,

$$\begin{aligned} & \sum_k \frac{(Mz)_k}{z_k} h_k^2 - h^T M h \\ &= \frac{1}{2} \sum_{k,\ell} m_{k\ell} \left(\sqrt{\frac{z_\ell}{z_k}} h_k - \sqrt{\frac{z_k}{z_\ell}} h_\ell \right)^2 \geq 0. \end{aligned}$$

We now consider the two block updates separately. For fixed V , let

$$C = V^T V, \quad E = X V,$$

and denote by $H = U^+ - U$ the change produced by (17). Since the graph term is independent of U ,

$$J_\theta(U^+, V) - J_\theta(U, V) = 2\langle UC - E, H \rangle + \text{Tr}(HCH^T).$$

Because C is symmetric and entrywise nonnegative, (21), applied to every row of U , gives

$$\text{Tr}(HCH^T) \leq \sum_{r,k} \frac{(UC)_{rk}}{u_{rk}} h_{rk}^2.$$

Moreover, update (17) implies

$$h_{rk} = -u_{rk} \frac{(UC - E)_{rk}}{(UC)_{rk}},$$

and

$$2(UC - E)_{rk} h_{rk} = -2 \frac{(UC)_{rk}}{u_{rk}} h_{rk}^2.$$

Therefore,

$$J_\theta(U^+, V) - J_\theta(U, V) \leq - \sum_{r,k} \frac{(UC)_{rk}}{u_{rk}} h_{rk}^2 \leq 0.$$

Next fix $U = U^+$. Set

$$A = U^T U, \quad B = X^T U,$$

and define

$$P = VA + \alpha D_\theta V, \quad Q = B + \alpha W_\theta V.$$

Let $G = V^+ - V$ and $G_{\mathcal{L}} = 0$, because the labeled rows remain fixed. Expanding the quadratic objective gives

$$J_\theta(U, V^+) - J_\theta(U, V) = 2\langle P - Q, G \rangle + \text{Tr}(GAG^T)$$

$$+\alpha \operatorname{Tr}(G^T L_\theta G). \quad (22)$$

Since A is symmetric and entrywise nonnegative, (21) yields

$$\operatorname{Tr}(GAG^T) \leq \sum_{i \in \mathcal{U}} \sum_k \frac{(VA)_{ik}}{v_{ik}} g_{ik}^2.$$

Furthermore, writing g_i for the i -th row of G , graph symmetry and nonnegativity give

$$\begin{aligned} \operatorname{Tr}(G^T L_\theta G) &= \frac{1}{2} \sum_{i,j} w_{\theta,ij} \|g_i - g_j\|_2^2 \\ &\leq \sum_{i,j} w_{\theta,ij} (\|g_i\|_2^2 + \|g_j\|_2^2) \\ &= 2 \sum_i d_{\theta,i} \|g_i\|_2^2 \\ &= 2 \sum_{i \in \mathcal{U}} \sum_k \frac{(D_\theta V)_{ik}}{v_{ik}} g_{ik}^2. \end{aligned}$$

By (18), we have

$$g_{ik} = -v_{ik} \frac{(P - Q)_{ik}}{P_{ik}}, \quad i \in \mathcal{U},$$

and

$$2(P - Q)_{ik} g_{ik} = -2 \frac{P_{ik}}{v_{ik}} g_{ik}^2.$$

Substituting these estimates into (22), we obtain that

$$\begin{aligned} &J_\theta(U, V^+) - J_\theta(U, V) \\ &\leq -2 \sum_{i \in \mathcal{U}} \sum_k \frac{P_{ik}}{v_{ik}} g_{ik}^2 + \sum_{i \in \mathcal{U}} \sum_k \frac{(VA)_{ik}}{v_{ik}} g_{ik}^2 \\ &\quad + 2\alpha \sum_{i \in \mathcal{U}} \sum_k \frac{(D_\theta V)_{ik}}{v_{ik}} g_{ik}^2 \\ &= - \sum_{i \in \mathcal{U}} \sum_k \frac{(VA)_{ik}}{v_{ik}} g_{ik}^2 \leq 0. \end{aligned}$$

Thus both block updates are nonincreasing, and

$$J_\theta(U^{(t+1)}, V^{(t+1)}) \leq J_\theta(U^{(t+1)}, V^{(t)}) \leq J_\theta(U^{(t)}, V^{(t)}).$$

Finally, graph validity gives $\operatorname{Tr}(V^T L_\theta V) \geq 0$, so $J_\theta(U, V) \geq 0$. Hence, the nonincreasing objective sequence is bounded below and therefore converges. This completes the proof. $\square$

Remark 2. Theorem 3 establishes convergence of the objective values for the exact updates under the stated strict-positivity condition; finite-precision behavior with denominator protection is examined experimentally.

Proposition 5 (Stationarity of positive fixed points). *Let (U^*, V^*) be a fixed point of the exact multiplicative updates with all free entries strictly positive and $V_{\mathcal{L}}^* = C_{\mathcal{L}}$. Then the gradient of J_θ vanishes with respect to U and the free block $V_{\mathcal{U}}$. Hence (U^*, V^*) is first-order stationary for the reduced problem with the labeled rows fixed.*

Proof. At a strictly positive fixed point, each multiplicative ratio in (17) and (18) equals one. Therefore

$$U^*((V^*)^T V^*) = X V^*,$$

and, on the free rows,

$$(V^*((U^*)^T U^*) + \alpha D_\theta V^*)_{\mathcal{U}} = (X^T U^* + \alpha W_\theta V^*)_{\mathcal{U}}.$$

These are precisely the zero-gradient conditions for U and $V_{\mathcal{U}}$ in (16). $\square$

4.4 Computational complexity

Let $|\mathcal{E}_p| = O(np)$ denote the number of edges in the common p -nearest-neighbor support. Constructing the two candidate graphs requires $O(mn^2)$ time and $O(np)$ sparse storage.

For each graph, the calibration stage performs one harmonic propagation solve in each of the F folds. If $\mathcal{S}(n, |\mathcal{E}_p|, c)$ denotes the cost of one sparse solve with c right-hand sides, the calibration cost is

$$O(2F \mathcal{S}(n, |\mathcal{E}_p|, c)).$$

The computation of Δ , s , and θ_κ costs only $O(F)$. In the proposed method, $F = 5$ and $\kappa = 1$ are fixed; the alternative values used in the sensitivity study are not part of a single execution.

After calibration, W_θ remains fixed. Each multiplicative iteration costs

$$O(mnc + npc + nc^2),$$

and T iterations require

$$O(T(mnc + npc + nc^2)).$$

Hence, the total cost is

$$O(mn^2 + 2F \mathcal{S}(n, |\mathcal{E}_p|, c) + T(mnc + npc + nc^2)).$$

CGC-GOCNMF therefore has the same factorization complexity as GOCNMF, with additional computation confined to the pre-factorization calibration stage. The same separation holds for CGC-GNMFLD: calibration adds the common pre-optimization cost, while the downstream factorization retains the parent GNMFLD iteration structure.

5 Experimental evaluation and analysis

5.1 Datasets and experimental protocol

We evaluated GNMFLD, ERDNMF, GOCNMF, and CGC-GOCNMF on six image datasets: PIE, YaleB, COIL20, COIL100, Optdigits, and MNIST. Table 2 summarizes the dataset statistics and the graph parameters used by GOCNMF and CGC-GOCNMF.

Each sample was normalized to unit Euclidean norm. In each trial, $\max\{2, \lfloor 0.1n_k \rfloor\}$ samples from class k were labeled and the remainder were evaluated. We used 20 paired trials with seeds 20260617–20260636; labeled subsets and initial factors were matched within each controlled comparison. The unified comparison and ablation used separately recorded initialization streams, so their CGC–GOC differences need not coincide numerically although each table is internally paired. Approximate five-fold stratification was

Table 2: Dataset characteristics and fixed graph settings used in the experiments.

Dataset	Samples n	Features m	Classes c	Labels/trial	Neighbors p	α
PIE	2856	1024	68	272	3	1000
YaleB	2414	1024	38	227	2	1000
COIL20	1440	1024	20	140	3	10
COIL100	7200	1024	100	700	3	10
Optdigits	5620	64	10	557	4	10
MNIST	6996	784	10	695	4	10

Notes. Labels/trial is the total number of labeled samples in each trial. The neighborhood size p and graph-regularization parameter α were fixed before the paired comparisons and were shared by GOCNMF and CGC-GOCNMF.

audited over all 600 dataset–seed–fold combinations, and every training fold retained every class.

GNMFLD and ERDNMF used 200 iterations, whereas GOCNMF and CGC-GOCNMF used 50. The GOCNMF parent comparison keeps the solver, initialization, and budget fixed; Section 5.5 applies the same controlled design to GNMFLD, changing only $W^{(0)}$ to W_θ . Both CGC instantiations use $F = 5$, $\kappa = 1$, $\varepsilon = 10^{-12}$ for multiplicative denominators, and the stated $10^{-10}I$ harmonic stabilization when required. No unlabeled ground-truth labels were used for parameter selection or calibration.

Assignments were obtained directly by row-wise arg max. ACC and NMI are reported as mean $\pm$ sample standard deviation over 20 trials. Two-sided paired Wilcoxon tests [24] with baseline-wise Holm correction were

interpreted together with paired effect sizes and win/tie/loss counts. Comparisons with GNMFLD and ERDNMF in the unified table provide method-level context; the two parent comparisons isolate the calibration step.

5.2 Unified comparison of four direct clustering methods

Table 3 compares GNMFLD, ERDNMF, GOCNMF, and CGC-GOCNMF under the same paired protocol. All methods produce cluster assignments directly from their learned representations, without K -means or an external classifier. ERDNMF parameters were fixed before the formal trials using public settings or labeled-only pilot validation.

Table 3: Clustering performance of four direct semi-supervised NMF methods on six datasets.

Dataset	Method	ACC (%)	NMI (%)
PIE	GNMFLD	93.27 $\pm$ 0.77	94.41 $\pm$ 0.39
	ERDNMF	77.22 $\pm$ 1.41	84.95 $\pm$ 0.98
	GOCNMF	93.93 $\pm$ 1.29	95.50 $\pm$ 0.77
	CGC-GOCNMF	93.93 $\pm$ 1.29^{†§}	95.50 $\pm$ 0.77^{†§}
YaleB	GNMFLD	61.85 $\pm$ 1.30	62.62 $\pm$ 1.04
	ERDNMF	24.63 $\pm$ 1.61	38.51 $\pm$ 0.91
	GOCNMF	73.28 $\pm$ 1.53	73.43 $\pm$ 1.07
	CGC-GOCNMF	73.28 $\pm$ 1.53 ^{†§}	73.43 $\pm$ 1.07 ^{†§}
COIL20	GNMFLD	88.70 $\pm$ 0.98	91.36 $\pm$ 0.68
	ERDNMF	68.34 $\pm$ 1.79	75.79 $\pm$ 1.67
	GOCNMF	91.63 $\pm$ 1.01	93.32 $\pm$ 0.73
	CGC-GOCNMF	93.66 $\pm$ 1.13^{††§}	94.78 $\pm$ 0.70^{††§}
COIL100	GNMFLD	81.61 $\pm$ 0.70	89.48 $\pm$ 0.28
	ERDNMF	54.90 $\pm$ 1.16	72.48 $\pm$ 0.76
	GOCNMF	85.02 $\pm$ 0.70	91.28 $\pm$ 0.39
	CGC-GOCNMF	87.60 $\pm$ 0.69^{††§}	92.99 $\pm$ 0.39^{††§}
Optdigits	GNMFLD	98.40 $\pm$ 0.14	95.93 $\pm$ 0.29
	ERDNMF	82.27 $\pm$ 0.97	71.14 $\pm$ 0.98
	GOCNMF	98.37 $\pm$ 0.13	95.88 $\pm$ 0.26
	CGC-GOCNMF	98.63 $\pm$ 0.12^{††§}	96.52 $\pm$ 0.26^{††§}
MNIST	GNMFLD	93.43 $\pm$ 0.28	85.53 $\pm$ 0.44
	ERDNMF	56.14 $\pm$ 1.74	45.81 $\pm$ 0.90
	GOCNMF	93.55 $\pm$ 0.27	85.73 $\pm$ 0.46
	CGC-GOCNMF	94.29 $\pm$ 0.23^{††§}	87.15 $\pm$ 0.40^{††§}

Values are mean percentages $\pm$ sample standard deviations over 20 paired trials on unlabeled samples. Bold values indicate the highest mean for each dataset and metric. [†], [‡], and [§] denote significant improvements over GOCNMF, GNMFLD, and ERDNMF, respectively.

CGC-GOCNMF achieves the highest reported mean ACC and NMI on COIL20, COIL100, Optdigits, and MNIST among the four evaluated direct methods. It exactly matches GOCNMF on PIE and shows no meaningful difference on YaleB. Under the reported fixed-iteration protocol, it also outperforms ERDNMF on all six datasets. The paired significance results are reported in Table 5, with the GOCNMF comparison isolating the effect of calibration most directly.

5.3 Common-stopping robustness and a recent reproducible baseline

Fixed iteration budgets can confound model quality with solver speed. We therefore performed an additional common-stopping experiment with GNMFLD, GOCNMF, CGC-GOCNMF, and the recent SNMFWLP method [7]. For every method, stopping was declared when the relative Frobenius change of the row-normalized representation was at most 10^{-4} for five consecutive iterations, after at least 20 iterations, with a common cap of 1000 iterations. Row

normalization makes the criterion invariant to representation scaling. The reconstruction or model objective was monitored separately and was not used to stop the solver.

The public SNMFWLP repository contains the core multiplicative updates but no complete experiment driver. We therefore provide a MATLAB R2019a integration wrapper implementing data normalization, graph construction, stratified labeled splits, direct row-wise assignment, the common stopping rule, checkpointing, and CSV output. Its coefficients were selected from $\{1, 10, 100\}^2$ by a disjoint labeled-only pilot split with seed 20260717; held-out labeled samples did not enter training, and no unlabeled ground-truth label was used. The selected (α, β) values were $(100, 100)$ for PIE and $(10, 100)$ for the other five datasets. Formal runs used seeds 20260617–20260619. The algebraically equivalent association $U(U^T XV)$ was used instead of $(UU^T)XV$ to avoid a dense feature-by-feature intermediate; this changes storage cost, not the update equation.

Table 4: Common-stopping robustness over three disjoint formal seeds. Values are mean ACC/NMI percentages $\pm$ sample standard deviation on unlabeled samples.

Dataset	GNMFLD	GOCNMF	CGC-GOCNMF	SNMFWLP (2026)
PIE	92.87 $\pm$ 1.03/94.25 $\pm$ 0.56	95.16 $\pm$ 0.97/96.24 $\pm$ 0.48	95.16 $\pm$ 0.97/96.24 $\pm$ 0.48	96.07 $\pm$ 0.66/96.69 $\pm$ 0.42
YaleB	61.01 $\pm$ 0.99/61.92 $\pm$ 0.56	72.66 $\pm$ 1.05/73.33 $\pm$ 0.57	72.69 $\pm$ 1.04/73.33 $\pm$ 0.57	61.52 $\pm$ 0.94/62.42 $\pm$ 0.33
COIL20	89.23 $\pm$ 1.36/91.64 $\pm$ 0.88	92.51 $\pm$ 0.51/93.91 $\pm$ 0.18	94.38 $\pm$ 1.42/95.50 $\pm$ 0.85	91.90 $\pm$ 1.09/93.76 $\pm$ 0.64
COIL100	81.69 $\pm$ 0.49/89.45 $\pm$ 0.14	86.47 $\pm$ 0.10/92.16 $\pm$ 0.14	89.42 $\pm$ 0.44/94.22 $\pm$ 0.17	86.84 $\pm$ 0.52/92.83 $\pm$ 0.29
Optdigits	98.49 $\pm$ 0.13/96.08 $\pm$ 0.26	98.36 $\pm$ 0.06/95.82 $\pm$ 0.10	98.60 $\pm$ 0.05/96.44 $\pm$ 0.08	98.70 $\pm$ 0.17/96.61 $\pm$ 0.39
MNIST	93.53 $\pm$ 0.52/85.68 $\pm$ 0.79	93.69 $\pm$ 0.50/85.96 $\pm$ 0.84	94.30 $\pm$ 0.37/87.21 $\pm$ 0.65	94.16 $\pm$ 0.57/87.10 $\pm$ 0.92

All 54 GNMFLD/GOCNMF/CGC-GOCNMF runs satisfied the common stopping rule and the independently monitored objective nonincrease audit. CGC-GOCNMF retains its parent-level fallback on PIE and YaleB and remains better than GOCNMF on COIL20, COIL100, Optdigits, and MNIST. SNMFWLP is competitive on PIE, Optdigits, and MNIST, but is not uniformly superior: CGC-GOCNMF has higher means on YaleB, COIL20, and COIL100. SNMFWLP reached the common stopping threshold in all 18 formal runs, while its monitored unconstrained objective was not monotone on five of the six datasets; these diagnostics and its 1000-iteration cap are reported rather than inter-

preted as a stationary-point convergence guarantee. Because this robustness experiment has three formal seeds, it supports solver-budget robustness but does not replace the 20-seed primary comparison.

5.4 Paired statistical analysis

Two-sided paired Wilcoxon signed-rank tests were applied to the 20 trial-wise differences between CGC-GOCNMF and each baseline. Holm correction was performed separately for each baseline over the 12 comparisons formed by six datasets and two metrics. No p -value was reported when all paired differences were exactly zero.

Table 5: Holm-adjusted p -values for CGC-GOCNMF versus the three baselines.

Dataset	vs. GOCNMF		vs. GNMFLD		vs. ERDNMF	
	ACC	NMI	ACC	NMI	ACC	NMI
PIE	—	—	0.0382	2.29×10^{-5}	2.29×10^{-5}	2.29×10^{-5}
YaleB	0.6346	0.6346	4.39×10^{-4}	2.29×10^{-5}	2.29×10^{-5}	2.29×10^{-5}
COIL20	4.41×10^{-4}	2.29×10^{-5}	2.29×10^{-5}	2.29×10^{-5}	1.77×10^{-4}	2.29×10^{-5}
COIL100	1.91×10^{-5}	1.91×10^{-5}	4.39×10^{-4}	2.29×10^{-5}	2.29×10^{-5}	2.29×10^{-5}
Optdigits	4.41×10^{-4}	1.91×10^{-5}	4.39×10^{-4}	2.29×10^{-5}	1.77×10^{-4}	2.29×10^{-5}
MNIST	4.41×10^{-4}	1.91×10^{-5}	4.39×10^{-4}	2.29×10^{-5}	2.29×10^{-5}	2.29×10^{-5}

Bold values are significant at the 0.05 family-wise level. A dash denotes an exact tie over all 20 paired trials. Holm correction was applied separately for each baseline over the 12 dataset–metric comparisons.

Relative to GOCNMF, CGC-GOCNMF is significant on COIL20, COIL100, Optdigits, and MNIST for both metrics, exactly tied on PIE, and not significant on YaleB. All comparisons with GNMFLD and ERDNMF are significant under the reported protocol. Table 6 complements adjust-

ed p -values with 10,000-resample seed-level bootstrap intervals and W/T/L counts. The largest parent gains occur on COIL20 and COIL100; the Optdigits ACC gain is small, PIE is an exact tie, and YaleB shows only one negligible loss among 20 trials.

Table 6: Paired changes of CGC-GOCNMF relative to GOCNMF. Values are mean percentage-point changes with seed-level percentile-bootstrap 95% confidence intervals over 20 paired trials (10,000 resamples; MATLAB random seed 20261009). W/T/L denotes wins, exact ties, and losses.

Dataset	ΔACC (95% CI)	ΔNMI (95% CI)	W/T/L (ACC; NMI)
PIE	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0/20/0; 0/20/0
YaleB	-0.002 [-0.007, 0.000]	-0.000 [-0.001, 0.000]	0/19/1; 0/19/1
COIL20	2.027 [1.696, 2.362]	1.467 [1.157, 1.776]	20/0/0; 19/0/1
COIL100	2.577 [2.467, 2.691]	1.703 [1.626, 1.778]	20/0/0; 20/0/0
Optdigits	0.260 [0.231, 0.287]	0.632 [0.572, 0.693]	20/0/0; 20/0/0
MNIST	0.740 [0.669, 0.813]	1.424 [1.289, 1.555]	20/0/0; 20/0/0

For reproducibility and comparison transparency, Table 7 reports the parameters and their provenance for GNMFLD and ERDNMF. All settings were fixed before the formal paired trials, without using ground-truth labels from the unlabeled samples. For YaleB and MNIST, the ERDNMF coefficient was selected by five-fold stratified valida-

tion on labeled samples only, using 20 fixed pilot seeds disjoint from the formal trials. Candidate values were $\{1, 10, 10^2, 10^3, 10^4\}$; candidates failing the objective non-increase audit were excluded. No unlabeled ground-truth labels were used.

Table 7: Parameters and provenance of the external baselines used in the unified comparison.

Dataset	GNMFLD				ERDNMF	
	p	α	β	Parameter provenance	α	Parameter provenance
PIE	5	10^5	10	Pre-specified	400	Original authors' code
YaleB	5	10^5	100	Pre-specified	1000	Labeled-only CV
COIL20	5	10^4	10	Pre-specified	800	Original authors' code
COIL100	5	10^4	10	Transferred from COIL20	1000	Original authors' code
Optdigits	5	10^5	10	Transferred from MNIST	600	Original authors' code
MNIST	5	10^5	10	Pre-specified	1000	Labeled-only CV

“Pre-specified” denotes settings fixed before the formal trials; “Transferred” denotes settings inherited from a related dataset; “Original authors' code” denotes parameters taken from the public ERDNMF implementation; and “Labeled-only CV” denotes validation using labeled samples only. Both baselines were run for 200 iterations. The transferred and pilot-selected settings are explicitly identified and are not presented as dataset-wise optimal parameters.

5.5 Transfer to a second graph-regularized parent model

The general formulation in Section 3 is not tied to the hard one-hot rows of GOCNMF. We therefore transferred the same CGC rule to GNMFLD, whose supervision is imposed through the label-discrimination term in (19). For every dataset and seed, GNMFLD and CGC-GNMFLD used the same labeled subset, U_0, V_0 , (p, α, β) values from Table 7,

and 200 multiplicative iterations; only $W^{(0)}$ was replaced by W_θ . Because unit-normalized data satisfy $\|x_i - x_j\|_2^2 = 2(1 - x_i^T x_j)$, the Euclidean and cosine-distance p -NN orderings coincide, so the $p = 5$ parent support is preserved. This transfer was executed as a separate paired experiment family with its own recorded initialization stream; consequently, rounded GNMFLD means can differ slightly from Table 3, while every GNMFLD/CGC-GNMFLD pair remains exactly matched.

Table 8: Transfer of CGC to GNMFLD over 20 paired trials. Changes are CGC-GNMFLD minus GNMFLD in percentage points.

Dataset	$\bar{\theta}$	ACC: parent $\rightarrow$ CGC	ΔACC	NMI: parent $\rightarrow$ CGC	ΔNMI	W/T/L
PIE	0.855	$93.29 \pm 0.78 \rightarrow 93.94 \pm 0.80$	+0.652	$94.41 \pm 0.39 \rightarrow 95.10 \pm 0.43$	+0.695	20/0/0
YaleB	0.907	$61.85 \pm 1.30 \rightarrow 64.75 \pm 1.39$	+2.899	$62.63 \pm 1.04 \rightarrow 65.31 \pm 1.01$	+2.684	20/0/0
COIL20	0.860	$88.70 \pm 0.98 \rightarrow 90.92 \pm 1.04$	+2.215	$91.37 \pm 0.68 \rightarrow 93.09 \pm 0.78$	+1.725	20/0/0
COIL100	0.944	$81.60 \pm 0.71 \rightarrow 86.17 \pm 0.59$	+4.567	$89.48 \pm 0.27 \rightarrow 92.53 \pm 0.31$	+3.058	20/0/0
Optdigits	0.875	$98.40 \pm 0.14 \rightarrow 98.64 \pm 0.12$	+0.237	$95.93 \pm 0.29 \rightarrow 96.50 \pm 0.26$	+0.566	20/0/0
MNIST	0.945	$93.43 \pm 0.28 \rightarrow 94.29 \pm 0.28$	+0.856	$85.53 \pm 0.44 \rightarrow 87.14 \pm 0.47$	+1.605	20/0/0

ACC/NMI entries are mean percentages $\pm$ sample standard deviations. W/T/L is identical for ACC and NMI. All 12 dataset–metric comparisons remain significant after exact two-sided paired Wilcoxon testing with Holm correction ($p_{\text{adj}} = 2.29 \times 10^{-5}$). No transfer trial produced an exact fallback.

GNMFLD keeps all representation rows free and uses a discrimination penalty, unlike the hard one-hot rows of GOCNMF. Yet the unchanged CGC rule improves both metrics in every paired trial; gains range from 0.237 to 4.567 points in ACC and from 0.566 to 3.058 points in NMI. This supports transfer beyond the GOCNMF supervision mechanism, without implying universal nondegradation: the formal guarantees remain exact parent recovery at $\theta = 0$ and controlled objective perturbation for $\theta > 0$.

5.6 Industrial image case study on steel surface defects

We further examined the calibration behavior on the NEU-CLS steel-surface defect database [25], which contains 1800

grayscale images from six defect categories. Each image was resized to 32×32 , vectorized, and normalized to unit Euclidean norm. No pretrained representation, external classifier, or K -means post-processing was used. This industrial-data case study tests whether the graph-calibration behavior persists outside the six benchmark datasets.

Label rates of 5%, 10%, and 20% per class were considered. For each rate, 20 paired trials used the same labeled samples and formal seeds. ACC, NMI, and ARI were evaluated only on the unlabeled samples. The parameters were fixed before the formal trials: $p = 5$ and $\alpha = 10$ for GOCNMF and CGC-GOCNMF; $p = 5$, $\alpha = 10^4$, and $\beta = 10$ for GNMFLD; and $\alpha = 10^3$ for ERDNMF. The ERDNMF coefficient was selected using labeled-only pilot validation, without using unlabeled ground-truth labels.

Table 9: Clustering performance on NEU-CLS at different label rates.

Label rate	Method	ACC (%)	NMI (%)	ARI (%)
5%	GNMFLD	32.34 ± 5.24	14.94 ± 2.91	7.74 ± 4.13
	ERDNMF	23.26 ± 2.61	7.09 ± 1.43	4.54 ± 0.91
	GOCNMF	32.78 ± 5.29	15.26 ± 2.81	8.09 ± 4.19
	CGC-GOCNMF	$33.86 \pm 4.51^{\dagger\ddagger}$	$16.50 \pm 2.44^{\dagger\ddagger}$	$8.72 \pm 3.97^{\dagger\ddagger}$
10%	GNMFLD	37.76 ± 3.72	17.37 ± 2.56	10.01 ± 3.31
	ERDNMF	24.12 ± 2.29	7.61 ± 1.97	4.89 ± 1.19
	GOCNMF	37.51 ± 3.75	17.20 ± 2.64	9.76 ± 3.38
	CGC-GOCNMF	$40.32 \pm 3.44^{\dagger\ddagger}$	$20.56 \pm 2.52^{\dagger\ddagger}$	$11.85 \pm 3.30^{\dagger\ddagger}$
20%	GNMFLD	42.44 ± 2.48	20.18 ± 2.12	12.72 ± 2.59
	ERDNMF	25.59 ± 2.13	8.05 ± 1.02	5.20 ± 0.59
	GOCNMF	42.05 ± 2.54	19.93 ± 2.13	12.31 ± 2.67
	CGC-GOCNMF	$46.90 \pm 2.38^{\dagger\ddagger}$	$25.94 \pm 2.27^{\dagger\ddagger}$	$16.31 \pm 2.83^{\dagger\ddagger}$

Values are mean percentages $\pm$ sample standard deviations over 20 paired trials on unlabeled samples. Bold values indicate the highest mean at each label rate. $\dagger$, $\ddagger$, and $\S$ denote significant improvements over GOCNMF, GNMFLD, and ERDNMF, respectively. Holm correction was applied separately for each baseline over the nine application comparisons.

Table 10: Calibration behavior on NEU-CLS over 20 trials.

Label rate	Mean θ	SD of θ	Exact fallbacks
5%	0.296	0.287	7/20
10%	0.646	0.183	0/20
20%	0.790	0.063	0/20

Table 9 shows that CGC-GOCNMF achieves the highest mean ACC, NMI, and ARI among the four evaluated direct methods at all three label rates. Its advantage over GOCNMF increases with the label rate, reaching 4.84, 6.01, and 4.00 percentage points in ACC, NMI, and ARI, respectively, at the 20% label rate. All comparisons with the three baselines remain significant after baseline-wise Holm correction. The main value of this experiment is the consistent calibration behavior across increasing label rates.

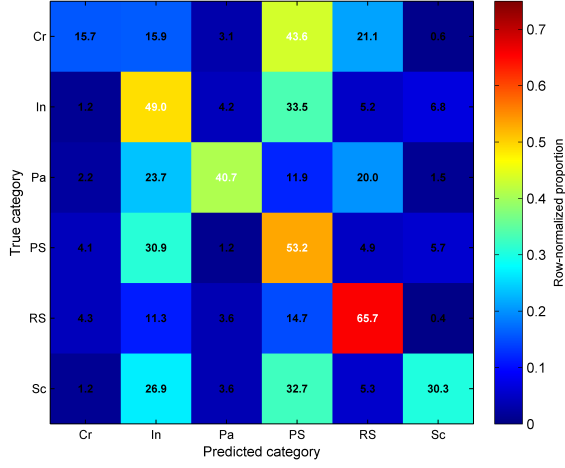
As shown in Table 10, limited labeled evidence produces stronger shrinkage and seven exact fallbacks at the 5% rate. As the label rate increases, θ rises and becomes more stable,

indicating more consistent support for the locally weighted graph.

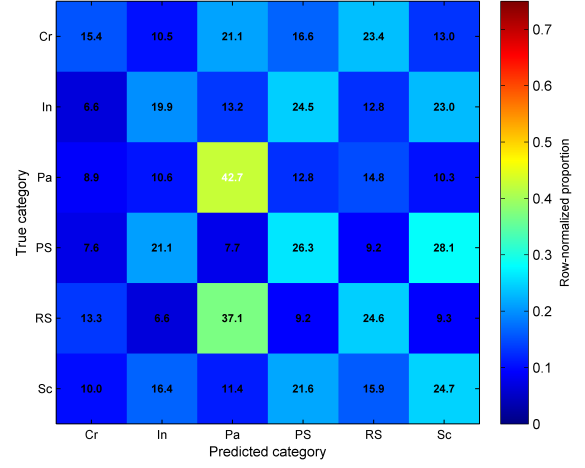
Figure 1 presents the aggregated confusion matrices at the 20% label rate. Relative to GOCNMF, CGC-GOCNMF improves the recalls of patches, rolled-in scale, and scratches, although crazing remains difficult when only raw intensity features are used.

5.6.1 Representation robustness

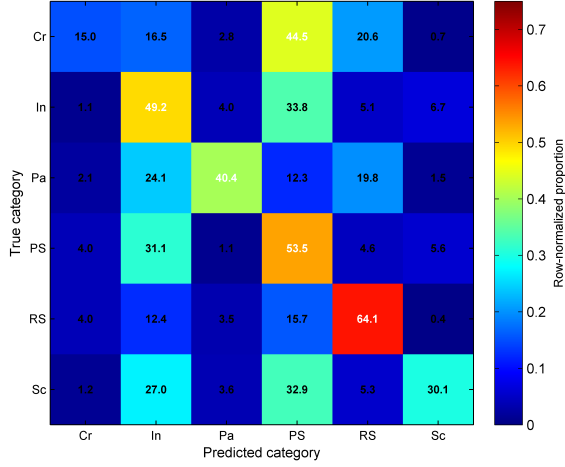
To test whether the calibration behavior depends on raw intensity vectors, we repeated the paired GOCNMF/CGC-GOCNMF comparison using a fixed 944-dimensional non-negative spatial uniform-LBP representation. Each image was resized to 34×34 , uniform LBP $_{8,1}^{u2}$ codes were computed on the valid 32×32 region, and normalized 59-bin histograms were extracted from a 4×4 cell grid. The label rates, 20 formal seeds, labeled subsets, initial factors, $p = 5$, $\alpha = 10$, 50 factorization iterations, five calibration folds, and direct row-wise assignment were retained.



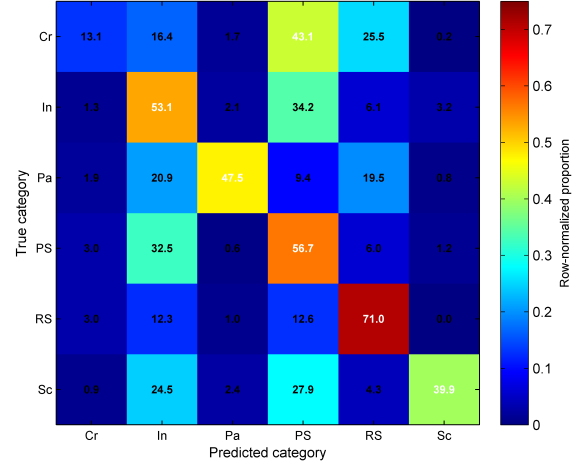
(a) GNMFLD



(b) ERDNMF



(c) GOCNMF



(d) CGC-GOCNMF

Figure 1: Aggregated row-normalized confusion matrices on NEU-CLS at the 20% label rate over 20 paired trials: (a) GNMFLD, (b) ERDNMF, (c) GOCNMF, and (d) CGC-GOCNMF. Rows denote true defect categories and columns denote predicted categories. Cr, In, Pa, PS, RS, and Sc denote crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches, respectively. Cell values are percentages, and all panels use the same color scale.

Table 11: Representation robustness on NEU-CLS using nonnegative spatial uniform-LBP features.

Label rate	Method	ACC (%)	NMI (%)	ARI (%)	$\bar{\theta}$	Fallbacks
5%	GOCNMF	27.42 ± 2.37	10.55 ± 1.86	3.19 ± 2.38	—	—
	CGC-GOCNMF	27.42 ± 2.37	10.55 ± 1.86	3.19 ± 2.38	0.000	20/20
10%	GOCNMF	32.66 ± 1.81	14.51 ± 1.13	5.58 ± 1.53	—	—
	CGC-GOCNMF	$33.38 \pm 2.19^*$	$15.81 \pm 1.51^*$	5.77 ± 1.59	0.357	8/20
20%	GOCNMF	38.53 ± 1.11	18.74 ± 1.40	9.39 ± 0.98	—	—
	CGC-GOCNMF	$41.39 \pm 1.20^*$	$22.76 \pm 1.25^*$	$10.91 \pm 1.03^*$	0.811	0/20

Values are mean percentages $\pm$ sample standard deviations over 20 paired trials on unlabeled samples. An asterisk denotes a significant improvement over GOCNMF after Holm correction over the nine label-rate-metric comparisons. Fallbacks count trials with $\theta = 0$.

At the 5% label rate, CGC-GOCNMF set $\theta = 0$ in all 20 trials and exactly reproduced GOCNMF. At the 10% rate, it improved ACC and NMI by 0.72 and 1.30 percentage points, respectively; both differences remained significant after Holm correction, whereas the ARI difference was not significant. At the 20% rate, the gains were 2.86, 4.03,

and 1.52 percentage points in ACC, NMI, and ARI, respectively, and all three differences were significant. Although the LBP representation did not improve the absolute clustering accuracy over raw pixels, the fallback and gain patterns remained coherent under a qualitatively different nonnegative texture representation.

5.7 Six-dataset ablation

We evaluated BASE, LOCAL, HARD, and SHRINK on all six benchmark datasets. All variants used the same graph

support, labeled subsets, initial factors, regularization coefficients, and iteration counts. BASE uses $W^{(0)}$, LOCAL uses $W^{(1)}$, HARD selects $W^{(1)}$ when $\Delta > 0$, and SHRINK uses W_θ .

Table 12: Six-dataset ablation of local graph weighting and conservative calibration.

Dataset	$\bar{\theta}$	Fallbacks	LOCAL			HARD			SHRINK		
			ΔACC	ΔNMI	Losses	ΔACC	ΔNMI	Losses	ΔACC	ΔNMI	Losses
PIE	0.000	20/20	-1.558	-0.761	20/20	0.000	0.000	0/0	0.000	0.000	0/0
YaleB	0.003	19/20	-4.088	-3.217	20/20	-0.428	-0.368	2/2	+0.005	+0.001	0/0
COIL20	0.770	0/20	+2.958	+2.171	0/0	+2.958	+2.171	0/0	+1.881	+1.420	0/0
COIL100	0.903	0/20	+2.709	+1.769	0/0	+2.709	+1.769	0/0	+2.472	+1.628	0/0
Optdigits	0.876	0/20	+0.305	+0.736	0/0	+0.305	+0.736	0/0	+0.255	+0.623	0/0
MNIST	0.943	0/20	+0.813	+1.573	0/0	+0.813	+1.573	0/0	+0.744	+1.434	0/0

Changes are absolute percentage points relative to BASE over 20 paired trials on unlabeled samples. “Losses” reports the ACC/NMI loss counts relative to BASE. Fallbacks count trials with $\theta = 0$.

Table 12 reveals two distinct regimes. LOCAL improves all 20 trials on COIL20, COIL100, Optdigits, and MNIST, but degrades both metrics in every PIE and YaleB trial. HARD removes the PIE failure but still selects the local graph in two YaleB trials, producing two ACC and two NMI losses. SHRINK exactly falls back in all PIE trials and in 19 of the 20 YaleB trials, while retaining substantial gains on the other four datasets. No ACC or NMI loss is observed for SHRINK in the 120 paired ablation runs, while the clear-benefit datasets retain most of the local-graph gain.

5.8 Sensitivity to the calibration rule

We generalized the calibration rule to

$$\theta_\kappa = \begin{cases} (\Delta - \kappa s)_+ / \Delta, & \Delta > 0, \\ 0, & \Delta \leq 0, \end{cases}$$

and considered $\kappa \in \{0, 0.5, 1, 1.5\}$ and $F \in \{3, 5, 10\}$. PIE, YaleB, and COIL100 were fixed before this experiment to represent fallback, borderline, and clear-benefit regimes, respectively. The labeled subsets and initial factors were paired across all settings.

Table 13: Sensitivity of conservative calibration to the shrinkage strength κ and the number of folds F .

Dataset	Setting	$\bar{\theta}$	Fallbacks	ΔACC	ΔNMI	Losses
<i>(a) Sensitivity to κ with $F = 5$</i>						
PIE	$\kappa = 0$	0.000	20/20	0.000	0.000	0/0
	$\kappa = 0.5$	0.000	20/20	0.000	0.000	0/0
	$\kappa = 1$	0.000	20/20	0.000	0.000	0/0
	$\kappa = 1.5$	0.000	20/20	0.000	0.000	0/0
YaleB	$\kappa = 0$	0.100	18/20	-0.428	-0.368	2/2
	$\kappa = 0.5$	0.027	19/20	-0.041	-0.041	1/1
	$\kappa = 1$	0.003	19/20	+0.005	+0.001	0/0
	$\kappa = 1.5$	0.000	20/20	0.000	0.000	0/0
COIL100	$\kappa = 0$	1.000	0/20	+2.709	+1.769	0/0
	$\kappa = 0.5$	0.952	0/20	+2.685	+1.774	0/0
	$\kappa = 1$	0.903	0/20	+2.472	+1.628	0/0
	$\kappa = 1.5$	0.855	0/20	+2.289	+1.504	0/0
<i>(b) Sensitivity to F with $\kappa = 1$</i>						
PIE	$F = 3$	0.000	20/20	0.000	0.000	0/0
	$F = 5$	0.000	20/20	0.000	0.000	0/0
	$F = 10$	0.000	20/20	0.000	0.000	0/0
YaleB	$F = 3$	0.030	19/20	-0.053	-0.053	1/1
	$F = 5$	0.003	19/20	+0.005	+0.001	0/0
	$F = 10$	0.001	19/20	+0.002	-0.001	0/1
COIL100	$F = 3$	0.925	0/20	+2.555	+1.679	0/0
	$F = 5$	0.903	0/20	+2.472	+1.628	0/0
	$F = 10$	0.908	0/20	+2.525	+1.665	0/0

Changes are absolute percentage points relative to BASE over 20 paired trials. “Losses” reports ACC/NMI loss counts. The proposed setting is $F = 5$ and $\kappa = 1$.

The qualitative behavior is stable across the tested settings. PIE always returns to BASE, whereas COIL100 retains a clear positive gain for every combination of F and κ . YaleB exposes the role of conservative shrinkage: with $F = 5$, hard selection ($\kappa = 0$) produces two losses, while increasing κ progressively contracts θ . The proposed choice

$\kappa = 1$ removes the observed losses while retaining a nonzero coefficient when supported; $\kappa = 1.5$ is more conservative and always falls back on YaleB. Varying F has little effect on PIE and COIL100, while the borderline YaleB case is most stable at $F = 5$ in these trials. Thus, $F = 5$ and $\kappa = 1$ provide a balanced conservative setting across the three tested

regimes.

$\rho \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$, define

$$W^{(\rho)} = (1 - \rho)W^{(1)} + \rho W^{(\text{perm})}.$$

5.9 Controlled graph-reliability stress test

We next tested whether the calibration reacts to controlled deterioration of the candidate graph. On COIL100, let $W^{(\text{perm})}$ be obtained by randomly permuting the undirected nonzero weights of $W^{(1)}$ over the same edge set. For

The construction preserves support, symmetry, nonnegativity, and the marginal edge-weight distribution while progressively destroying the association between edge location and local similarity. Performance is reported as paired change relative to BASE.

Table 14: Graph-reliability stress-test results on COIL100.

ρ	Edge corr.	$\bar{\theta}$	$\bar{\Delta}$	ACC			NMI		
				Candidate	SHRINK	Losses	Candidate	SHRINK	Losses
0.0	+1.000	0.903	+0.05736	+2.709	+2.472	0/0	+1.769	+1.628	0/0
0.2	+0.970	0.923	+0.03426	+2.004	+1.753	0/0	+1.327	+1.162	0/0
0.4	+0.832	0.915	+0.02133	+1.217	+1.096	0/0	+0.790	+0.716	0/0
0.6	+0.555	0.869	+0.01185	+0.508	+0.455	0/1	+0.329	+0.302	0/0
0.8	+0.243	0.489	+0.00377	-0.168	+0.074	14/5	-0.127	+0.054	14/3
1.0	-0.000	0.018	-0.00454	-1.125	+0.002	20/0	-0.767	+0.002	20/0

Results are based on 20 paired trials. Performance changes are absolute percentage points relative to BASE. “Losses” reports the candidate/SHRINK loss counts.

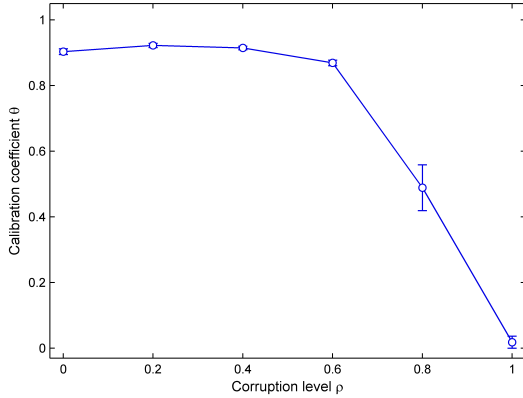


Figure 2: Response of the calibration coefficient to controlled graph corruption on COIL100. Points are means over 20 paired trials, and error bars denote standard errors.

As ρ increases, the mean edge-weight correlation decreases from 1 to approximately 0, while the mean out-of-fold advantage decreases monotonically from 0.05736 to -0.00454 . The uncalibrated candidate changes from a clear improvement to a consistent loss. Although $\bar{\theta}$ is not strict-

ly monotone because it depends on s_ρ/Δ_ρ , it decreases to 0.489 at $\rho = 0.8$ and 0.018 at $\rho = 1$; at $\rho = 1$, 19 of the 20 trials yield exact fallback.

At $\rho = 0.8$, the candidate loses ACC and NMI in 14 trials, whereas SHRINK reduces these counts to 5 and 3. At $\rho = 1$, the candidate loses both metrics in all 20 trials, while SHRINK records no loss and is practically identical to BASE. These results show that the proposed calibration responds conservatively to severe graph deterioration and progressively returns the downstream model toward the baseline behavior.

5.10 Prescribed-protocol training time

We measured the prescribed-protocol training time of each method over 20 runs. Data loading, common feature normalization, and post-training evaluation were excluded. For the graph-based methods, the measured time included graph construction and factorization. CGC-GOCNMF additionally included the complete five-fold calibration procedure. ERD-NMF has no graph or calibration stage, so its reported time consists only of factorization. An untimed warm-up was performed before the formal runs.

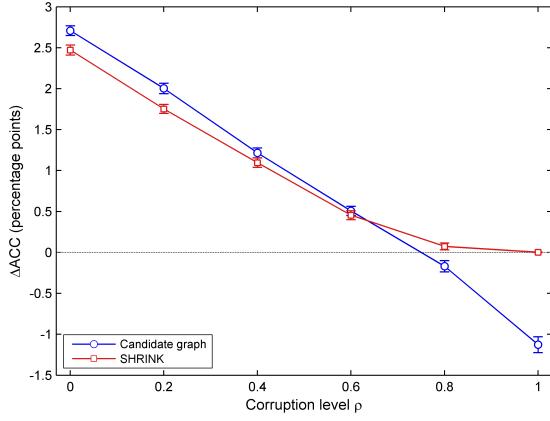
Table 15: Prescribed-protocol training times of the four direct methods in seconds.

Dataset	GNMFLD	ERDNMF	GOCNMF	CGC-GOCNMF	T_{cal}	Overhead (%)	Cal. share (%)
PIE	10.719 $\pm$ 0.490	19.017 $\pm$ 0.091	4.068 $\pm$ 0.198	4.504 $\pm$ 0.426	0.410	10.73	9.09
YaleB	5.981 $\pm$ 0.626	12.049 $\pm$ 1.581	2.385 $\pm$ 0.251	2.407 $\pm$ 0.209	0.050	0.95	2.10
COIL20	3.057 $\pm$ 0.036	5.236 $\pm$ 0.073	1.133 $\pm$ 0.016	1.172 $\pm$ 0.024	0.024	3.41	2.07
COIL100	49.974 $\pm$ 7.103	58.937 $\pm$ 9.199	25.014 $\pm$ 3.619	28.542 $\pm$ 3.892	3.704	14.11	12.98
Optdigits	8.131 $\pm$ 0.437	1.875 $\pm$ 0.028	7.720 $\pm$ 0.344	7.872 $\pm$ 0.152	0.176	1.98	2.24
MNIST	22.143 $\pm$ 4.320	24.411 $\pm$ 2.509	17.492 $\pm$ 3.066	18.571 $\pm$ 3.548	0.436	6.17	2.35

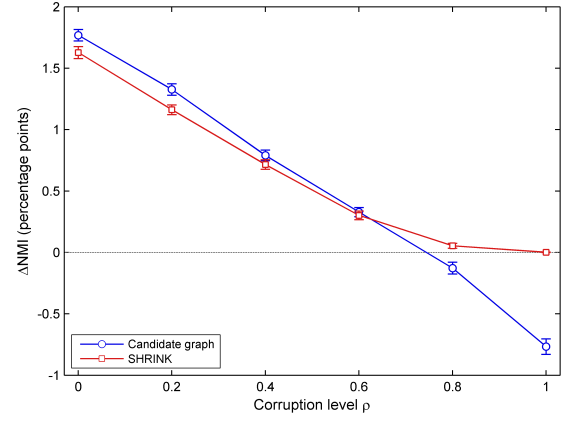
T_{cal} is the CGC-GOCNMF calibration time. “Overhead” is the relative increase over GOCNMF, while “Cal. share” is the percentage of calibration time in the total CGC-GOCNMF time.

As shown in Table 15, the additional cost of CGC-GOCNMF over GOCNMF ranges from 0.95% to 14.11%. Calibration accounts for at most 12.98% of its total running time. Because the methods use different objectives and itera-

tion counts, only the runtime difference between GOCNMF and CGC-GOCNMF is directly interpretable; the remaining values are protocol-level references.



(a) ACC



(b) NMI

Figure 3: Performance changes relative to BASE under controlled corruption. The uncalibrated candidate deteriorates as ρ increases, whereas SHRINK substantially reduces the losses.

5.11 Discussion and limitations

The experiments reveal complementary calibration regimes rather than a requirement that the candidate graph always be used. In the primary GOCNMF instantiation, stable positive evidence retains substantial local weighting on COIL20, COIL100, Optdigits, and MNIST; weak evidence contracts the coefficient on YaleB; and insufficient evidence produces exact fallback on PIE and in the low-label LBP experiment. The controlled corruption study strengthens this mechanism-level interpretation by showing that deliberate deterioration of edge-location information reduces the estimated candidate advantage and drives the downstream model back toward the baseline.

The GNMFLD transfer addresses a different question: whether this behavior is tied to GOCNMF’s hard one-hot supervision. GNMFLD instead uses a label-discrimination penalty, yet the unchanged CGC rule improves both metrics in all 20 paired trials on all six benchmarks. Together, the two instantiations support the intended separation between the calibration stage and the downstream supervision mechanism. Propositions 1–3 explain this separation structurally: θ controls graph, Laplacian, and downstream graph-regularized objective deviation, while Theorem 2 provides exact parent recovery when the retained evidence vanishes.

Three limitations remain. First, the variability discount is deterministic rather than a calibrated confidence procedure. Second, both downstream instantiations are NMF models and the datasets are predominantly image-based; non-NMF Laplacian-regularized learners and non-image data would test broader transfer. Third, the common-stopping and SNMFWLP robustness study uses three formal seeds, whereas the primary fixed-budget study uses 20; a 20-seed common-stopping replication would permit equally powered inferential comparisons.

6 Conclusion

This paper introduced evidence-guided Conservative Graph Calibration as a pre-optimization mechanism for Laplacian-regularized semi-supervised representation models. Out-of-

fold label propagation evaluates a prescribed baseline graph and a same-support locally weighted candidate, while a variability-adjusted retained-advantage coefficient controls how far the final graph departs from the baseline. The analysis shows that this coefficient scales graph and Laplacian deviation, bounds the induced graph-regularized objective perturbation, and yields exact parent recovery when the evidence for modification is insufficient.

The primary CGC-GOCNMF instantiation improves its parent on COIL20, COIL100, Optdigits, and MNIST and retains essentially baseline behavior on PIE and YaleB. The same calibration rule transfers without retuning to GNMFLD, a structurally different parent using label-discrimination supervision: CGC-GNMFLD improves ACC and NMI in all 20 paired trials on each of the six benchmarks, with mean ACC gains of 0.24–4.57 percentage points and NMI gains of 0.57–3.06 points. Industrial-image, representation-robustness, sensitivity, ablation, and controlled-corruption experiments provide complementary evidence about when local weighting is retained, attenuated, or rejected. Taken together, the two parent-model studies support the view that limited labels can be used not only to supervise representation learning but also to regulate structural graph decisions before optimization.

CRedit authorship contribution statement

Deshu Sun: Writing-original draft, Methodology, Software, Conceptualization, Formal analysis. Feng Wang: Writing-review and editing, Funding acquisition, Visualization, Conceptualization. All authors reviewed the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The MATLAB R2019a code, trial-level outputs, summary files, figures, manuscript source, and the redistributable Optdigits matrix are available at <https://github.com/matrix-research-lab-Wf/CGC-NMF>. The repository records exact matrix dimensions and SHA-256 checksums for every dataset used. PIE, YaleB, COIL20, COIL100, MNIST, and NEU-CLS matrices are not redistributed where an explicit compatible redistribution license could not be verified; authoritative access links and local preparation instructions are provided instead.

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